INNOVATIVE CATALYSTS FOR AFFORTABLE GREEN HYDROGEN PRODUCTION.

Introduction

The need for an alternative, renewable, and sustainable solution to the energy and water crisis is becoming increasingly important due to the continuous increase in industrial activities caused by population growth. The energy sector is facing serious challenges to sustain the production of everyday necessities such as transportation, agriculture, and clean water resources. In recent years, green hydrogen have been identified as a possible solution to mitigate the current global issues. The challenges with this innovative technology arise due to the expensive costs associated with splitting water molecules into molecular oxygen and hydrogen gas using water electrolysis techniques. Commercially, three main types of water electrolysers are available. These include proton exchange membrane (PEM) electrolysers, solid oxide electrolyser cells (SOEC), and anion exchange membrane (AEM) electrolyser systems. Each technology has its benefits and limitations; however, the main challenge is costs related to noble metal catalysts and materials required to maintain an effective rate of green hydrogen production.

This study presents a circular and solar-integrated approach combining phytoremediation, biochar-based catalysts, and an AEM water electrolyser system to generate green hydrogen. This approach incorporates heavy metals, transition metals, and platinum group metals into active catalytic materials that can be used in the AEM system. Since a renewable energy source such as solar energy is used to convert the electrical energy to chemical energy in the form of hydrogen-based fuels, energy costs is reduced and the only contribution to greenhouse gases comes from the preparation of the biochar-catalysts. This significantly reduces operational costs, minimizes pollution, provides a new clean water source, and lowers the environmental impact since the hydrogen produced can be used to generate downstream chemicals such as fertilizers for countries that suffer from a lack of infrastructure to support agricultural activities.

Problem Statement

The environment and surrounding communities are often the main factors that many industrial companies ignore when setting up manufacturing plants. The waste generated by these companies are often discarded into the environment, which consequently poses significant threats to surrounding ecosystems and human health. Many industrial effluents contain toxic chemicals that pollute waterways used by communities. There is a need for an integrated, sustainable solution that can recover and reuse toxic heavy metals from wastewater while simultaneously generating green hydrogen for commercial use. Bridging these challenges requires a circular process capable of achieving pollution remediation, material valorization, and cost-effective energy generation.

Objective of the Study

To develop and validate a circular system that converts metal-contaminated biomass into efficient, low-cost biochar-based catalysts for AEM hydrogen production.

Scope of the Study

This study focuses on integrating phytoremediation, biochar synthesis, and AEM electrolysis withing a unified framework. It explores the selection of suitable plant species for metal absorption, the conversion of harvested biomass into conductive biochar, and the synthesis of Ru-biochar catalysts. Laboratory tests assess catalyst performance, hydrogen production yield, and overall system efficiency. Environmental and techno-economic analyses are included to evaluate the viability of the system. However, large-scale industrial application remains outside the study's scope and will be addressed in future pilot-scale demonstrations.

Methodology/Experimental

The methodology includes literature review, laboratory experiments, and techno-economic analysis. Key steps involve identifying efficient hyperaccumulator plants, optimizing biochar production through pyrolysis, and synthesizing Ru-biochar catalysts. Electrochemical testing is performed to evaluate hydrogen production efficiency using AEM electrolysis. The study also employs cost modeling and life-cycle assessment to quantify economic and environmental benefits. Collaboration with experts in electrochemistry, materials science, and environmental engineering ensures that the findings are scientifically robust and aligned with real-world applicability.